

LAB 5-CS2610
CS23B096-ATHARVA MOGHE

Q1)

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core 0: >>>> user_mode
core 0: 0x0000000080000076 (0x00004505) c.li    a0, 1
(spike)
core 0: 0x0000000080000078 (0x0000458d) c.li    a1, 3
(spike)
core 0: 0x000000008000007a (0x0000461d) c.li    a2, 7
(spike)
core 0: 0x000000008000007c (0x00000073) ecall
core 0: exception trap_user_ecall, epc 0x000000008000007c
(spike)
core 0: >>>> supervisor_trap
core 0: 0x0000000080000080 (0x00000073) ecall
core 0: exception trap_supervisor_ecall, epc 0x0000000080000080
(spike)
core 0: >>>> machine_trap
core 0: 0x0000000080000094 (0x341022f3) csrr    t0, mepc
(spike)
core 0: 0x0000000080000098 (0x00000291) c.addi  t0, 4
(spike)
core 0: 0x000000008000009a (0x34129073) csrw    mepc, t0
(spike)
core 0: 0x000000008000009e (0x30200073) mret
(spike)
core 0: 0x0000000080000084 (0x00000297) auipc  t0, 0x0
(spike)
core 0: 0x0000000080000088 (0xff228293) addi    t0, t0, -14
(spike)
core 0: 0x000000008000008c (0x14129073) csrw    sepc, t0
(spike)
core 0: 0x0000000080000090 (0x10200073) sret
(spike)
core 0: >>>> user_mode
core 0: 0x0000000080000076 (0x00004505) c.li    a0, 1
(spike) □
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The code follows the following pipeline-

User mode -> supervisor trap -> machine trap -> supervisor trap(denoted by sret on line 0x90)
->user mode

Also attaching screenshots of the first iteration where pipeline is
machine_mode->supervisor_mode->user_mode

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core  0: >>>> machine_mode
core  0: 0x0000000080000000 (0x00000297) auipc    t0, 0x0
(spike)
core  0: 0x0000000080000004 (0x04428293) addi     t0, t0, 68
(spike)
core  0: 0x0000000080000008 (0x34129073) csrw     mepc, t0
(spike)
core  0: 0x000000008000000c (0x00006289) c.lui    t0, 0x2
(spike)
core  0: 0x000000008000000e (0x8002829b) addiw    t0, t0, -2048
(spike)
core  0: 0x0000000080000012 (0x30002373) csrr     t1, mstatus
(spike)
core  0: 0x0000000080000016 (0xffff2c393) not      t2, t0
(spike)
core  0: 0x000000008000001a (0x00737333) and      t1, t1, t2
(spike)
core  0: 0x000000008000001e (0x00006e05) c.lui    t3, 0x1
(spike)
core  0: 0x0000000080000020 (0x800e0e1b) addiw    t3, t3, -2048
(spike)
core  0: 0x0000000080000024 (0x01c36333) or       t1, t1, t3
(spike)
core  0: 0x0000000080000028 (0x30031073) csrw     mstatus, t1
(spike)
core  0: 0x000000008000002c (0x00000f97) auipc    t6, 0x0
(spike)
core  0: 0x0000000080000030 (0x068f8f93) addi     t6, t6, 104
(spike)
core  0: 0x0000000080000034 (0x305f9073) csrw     mtvec, t6
(spike)
core  0: 0x0000000080000038 (0x00004f05) c.li     t5, 1
(spike)
core  0: 0x000000008000003a (0x00000f22) c.slli   t5, 8
(spike)
core  0: 0x000000008000003c (0x302f2073) csrs     medeleg, t5
(spike) □

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core 0: >>>> supervisor_mode
core 0: 0x0000000080000044 (0x00000e97) auipc    t4, 0x0
(spike)
core 0: 0x0000000080000048 (0x03ce8e93) addi     t4, t4, 60
(spike)
core 0: 0x000000008000004c (0x105e9073) csrw     stvec, t4
(spike)
core 0: 0x0000000080000050 (0x00000297) auipc    t0, 0x0
(spike)
core 0: 0x0000000080000054 (0x02628293) addi     t0, t0, 38
(spike)
core 0: 0x0000000080000058 (0x14129073) csrw     sepc, t0
(spike)
core 0: 0x000000008000005c (0x00006289) c.lui    t0, 0x2
(spike)
core 0: 0x000000008000005e (0x8002829b) addiw    t0, t0, -2048
(spike)
core 0: 0x0000000080000062 (0x10002373) csrr     t1, sstatus
(spike)
core 0: 0x0000000080000066 (0xffff2c393) not     t2, t0
(spike)
core 0: 0x000000008000006a (0x00737333) and      t1, t1, t2
(spike)
core 0: 0x000000008000006e (0x10031073) csrw     sstatus, t1
(spike)
core 0: 0x0000000080000072 (0x10200073) sret
(spike)
core 0: >>>> user_mode
core 0: 0x0000000080000076 (0x00004505) c.li     a0, 1
(spike)
core 0: 0x0000000080000078 (0x0000458d) c.li     a1, 3
(spike) □

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